

— UCP-LEANING — NDP-LEANING

The Quiet Part.

Alberta's electoral-boundary commission split three to two. The government threw out both maps and handed the drafting pencil to five MLAs instead. An audit of what was rejected, what the government is promising, and what to watch for in November.

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AUDIT EXECUTIVE SUMMARY

github.com/Ixby/alberta-electoral-boundaries-audit

COVER MAP • The 2026 minority commission proposal — the map this audit ends up critiquing. Each district is coloured by its 2023 UCP-NDP vote share.

Two Maps, Then None: Inside Alberta's 2026 Boundary Audit

A plain-language look at the 2025–26 Electoral Boundary Commission, the math behind the minority map, and what comes next.

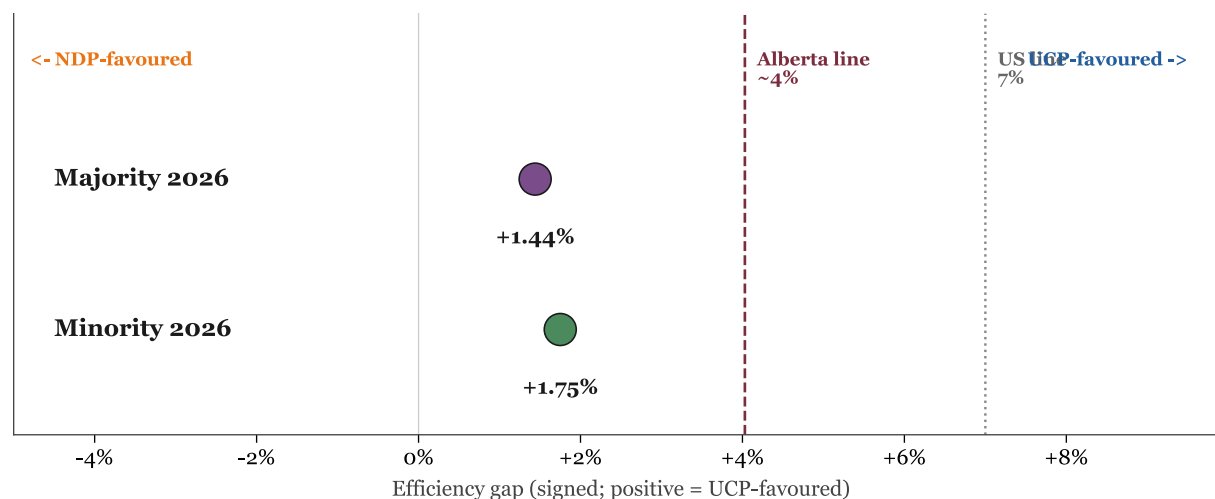
Part I: How the Commission Broke

Alberta's Electoral Boundary Commission finished its work on March 23, 2026 and could not agree. Three commissioners produced one map; the other two produced a different one. Both are legal under the *Electoral Boundaries Commission Act*. The governing party is the United Conservative Party (UCP); its main opposition is the New Democratic Party (NDP). This audit measured both maps using the same methods, applied identically. Three findings stand out.

1. **The two maps differ on six things you can measure without looking at any election results:** how evenly people are spread across districts, whether voters are concentrated, how badly cities are cut up, whether borders follow city limits, the shape of the districts, and how many boundaries the commission's own chair flagged as anomalous. The minority map differs from the majority on every one of them.
2. **The direction runs against the places where the governing party is weakest.** Calgary's northwest quadrant, the City of Airdrie, urban areas with established municipal boundaries — the communities most affected by the minority map's structural differences are the same ones where the governing party's opponents are strongest. The audit cannot determine intent. It can measure effect.
3. **The process now promoting the minority map has no precedent in Canada.** No other province lets a cabinet hand redistricting to a committee its own party controls partway through a redistribution cycle. Most provinces either require the legislature to debate the commissioners' map first, or give the commission's map automatic effect unless overridden. Alberta does neither. On April 16, the government set both commission maps aside and assigned the work to a five-member committee of MLAs (Members of the Legislative Assembly), three from the governing United Conservative Party (UCP).

The process is its own finding, separate from the maps.

Part II: The 250,000-Map Litmus Test



The table compares the two maps. The first five rows use no election results — they're properties of the lines themselves. The last two depend on how votes were attributed to each district.

WHAT WAS MEASURED	MAJORITY MAP	MINORITY MAP
Population spread across districts (tighter is better)	3,180	4,707 — 48% wider
NW Calgary population excess above average	2.8%	11.5%
Airdrie split	2 divisions	4 divisions
Borders that follow existing municipal lines	71%	15%
Boundaries flagged by the commission chair	0	3
Seats at 50/50 votes (percentile in 250k simulation)	46.1% — p77 (normal range)	48.3% — p98.6 (top 1.5%)
Efficiency gap (a partisan-fairness measure)	+1.4%	+1.8%
Packing-cracking neighbourhood pattern	3 (same as 2019)	0 — pre-registered pass

VOCABULARY

EFFICIENCY GAP.

A single number that measures how lopsidedly a party's votes are translated into seats. Positive numbers favour conservatives; negative favour the NDP. The audit uses ~5% as Alberta's outlier line — roughly the level only 5% of computer-simulated Alberta maps cross.

ANCHORING.

The fraction of an electoral border that lies on a pre-existing administrative line — a city limit, a school-division boundary, a Statistics Canada census line.

The bottom rows depend on election results. The *seats@50/50* test holds the electorate at perfect parity (UCP and NDP each win exactly half the votes province-wide) and asks how many seats the map awards the UCP. A neutral Alberta map produces a median around 44.8% UCP seats — Alberta's geography (NDP voters concentrated in city cores, UCP voters spread across rural ridings) gives the NDP a small efficiency advantage at neutrality. The majority map at 46.1% sits at the 77th percentile of the 250,000-map simulation (normal range). The minority map at 48.3% sits at the 98.5th percentile (top 1.5%). The *efficiency gap* number measures how lopsidedly each party's votes get translated into seats; on the corrected v0_9 substrate both maps' efficiency gaps are inside the simulation's bulk (+1.4% and +1.8%) and so do not separate the two maps the way *seats@50/50* does. The verdict section unpacks the consequences.

The last row is where the minority map looks *better* than the majority on a partisan-fairness measure: zero packed-cracked adjacency pairs, against three on the majority and three on the current 2019 map. The audit pre-registered this test before measuring, and the minority's tighter packing-cracking number is a real point in its favour. It is the single test where the minority outperforms the majority.

Part III: Cracking, Packing, and Draining

Airdrie: one city, four partitions

Population ~90,000. The majority draws two ridings both named Airdrie. The minority cuts the city into four pieces — one per compass direction — each attached to a different surrounding district.

Majority proposal · 2 ridings, both named Airdrie



Minority proposal · 4 segments, only one named Airdrie



The minority cuts Airdrie into four pieces, each attached to a different surrounding district. Only one of the four is named Airdrie alone; the other three carry Calgary or regional names. The majority draws two compact ridings, both named Airdrie.

Source: commission majority report Appendix A and minority report Appendix E.

The five commissioners worked from the same statutory rules, the same provincial geography, the same archive of 1,140 public submissions, and the same demographic data. Their two competing drafts agree on most of Alberta. Where the drafts diverge, they diverge on choices someone in the room had to make. Three of those choices are worth seeing as choices, not numbers.

It splits the City of Airdrie into four pieces. The law caps each electoral division at one-and-a-quarter times the provincial average, so Airdrie needs at least two divisions. The majority map gives it two. The minority gives it four — north to Calgary-Nolan Hill-Cochrane, east to Airdrie East, west to Calgary-Foothills-Airdrie West, and centre-south to Calgary-Airdrie — each one stapled to a different rural or Calgary-edge district. An Airdrie resident with a question for her MLA has to know which quarter of

the city she lives in before she can call the right office. Her neighbours two blocks over will give her three different answers. The PTA at her child's school cannot send a single delegation to one MLA on a school-funding question; they have to coordinate four delegations to four offices, each MLA primarily accountable to a different rural or suburban constituency. The minor-hockey association, the food bank, the Chamber of Commerce — every organization that operates citywide now operates across four provincial ridings.

It ignores municipal boundaries. When electoral maps follow the edge of a city or town, voters recognize where their division begins and ends — the property-tax line, the school-division line, the local-election ward line, and the provincial-election line all coincide. These maps already exist. Statistics Canada publishes the city and town boundaries; any redistricting commission gets them for free. The majority map traces them on 71 of every 100 kilometres of its border. The minority traces them on 15 of 100 — and offers no published reason for the departure. (For comparison, recent Quebec, Ontario, and BC commission maps anchor at 78%, 82%, and 71% respectively. Canada's 70–85% norm comes from these and seven other comparator commissions documented in the monograph.)

One area of Calgary is carved up to concentrate NDP voters into larger-than-average divisions. In Calgary's northwest quadrant, the minority map's divisions average 11.5% above the province-wide population — versus 2.8% on the majority. The same geographic zone, drawn by the same commission under the same constraints, produces districts a quarter larger on one map than on the other. This is *packing*: concentrating one party's voters into fewer, larger districts so each of their ballots weighs less. Packing and *cracking* (splitting a party's voters thinly across districts they narrowly lose) are the two classic gerrymandering moves; both shrink a party's seat count below its vote share.

The commission chair — appointed under the same Act, working from the same submissions — flagged three boundaries on the minority map as geographically anomalous: Rocky Mountain House–Banff Park's extension into uninhabited national park land; the Calgary–Nolan Hill–Cochrane lasso-shaped corridor; the Olds–Three Hills–Didsbury reach into north Airdrie. The majority received zero such flags from the same chair.

WHY AIRDRIE MATTERS

Airdrie is the largest Alberta city without its own MLA. At 84,000 people it is bigger than Red Deer; it has one council, one tax bill, one school division — every civic system treats it as a unit.

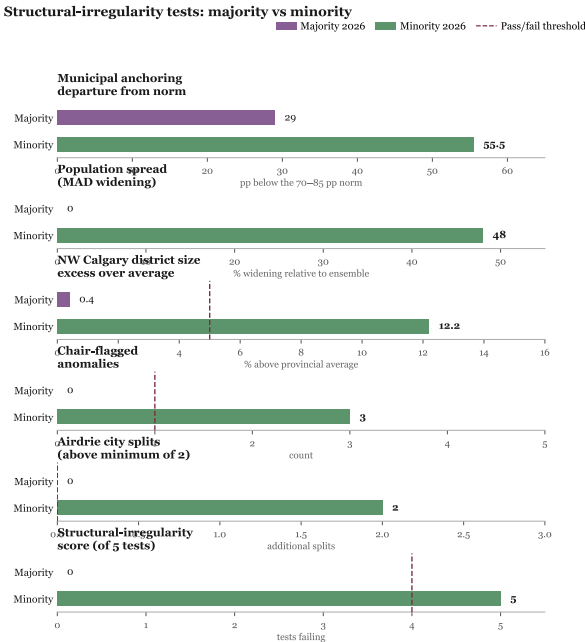
Splitting it across four provincial divisions — Calgary–Airdrie, Calgary–Foothills–Airdrie West, Calgary–Nolan Hill–Cochrane, and Airdrie East — each primarily identified with a different surrounding jurisdiction, removes Airdrie from the political map at the level of government that draws it. The city has 84,000 residents and zero seats in the legislature where a majority of voters call the place home.

A four-way split is invisible to every partisan-fairness test except the one that asks: can a voter find their MLA?

Both maps are legal. The four-way split is a choice.

Part IV: The Impact on the Ground

Lane 1 depends on which election results you score the maps against. Lane 2 does not. The structural evidence is in the maps themselves — drawn lines, split cities, where the boundaries do and don't follow administrative lines that exist for other reasons. On these tests, the two maps are not close.



The same five tests in tabular form, with each test's threshold stated alongside the result. The bottom row is the audit's

summary

— the count of tests each map fails out of the five.

TEST	MAJORITY MAP	MINORITY MAP
Border follows existing municipal lines (70–85% Canadian norm)	71% — within norm	15% — 4.9× below norm
Population spread (tighter is better)	3,180	4,707 — 48% wider
NW Calgary population excess above average	2.8%	11.5%
Boundaries flagged by the commission's own chair	0	3
Airdrie split (constraint minimum: 2)	2 pieces	4 pieces
Pre-registered summary (4 of 5 fired = outlier)	0 of 5 fired	5 of 5 fired

A separate finding — applied only to the minority because the minority is the map whose contested redraws are public¹ — is that **five of six of the minority commissioners' published rationales fail under independent check**. (A seventh redraw the audit had previously listed turned out to rest on a federal-boundary claim that cannot be located in the minority report; it has been removed rather than left as a weak claim.)

The audit also tested the chair's separate, blanket assertion in Appendix C that the minority's seven contested hybrid configurations had **no public support** in the 1,140+ public submissions. A keyword

search across the full submission archive (94% machine-parsed, 6% image-only and excluded; methodology and per-configuration evidence at [analysis/reports/submission_search_findings.md](#)) returned a more nuanced picture than either the chair's blanket claim or its blanket dismissal: the chair was right on three of seven (Airdrie 4-way split, Calgary–Nolan–Hill–Cochrane hybrid, and the St. Albert minority alternative each lack any documented support), wrong on three of seven (Rocky Mountain House–Banff Park drew an explicit, detailed proposal from at least one Clearwater-area submission plus several aligned ones; Olds–Three–Hills–Didsbury was supported by Beiseker residents in writing; Chestermere drew multiple submissions opposing a Calgary merger that materially align with the minority's intent), and partially wrong on one (Red Deer hybrids drew a peri-Red-Deer hybrid proposal from a sitting Red Deer councillor, with directional but not configuration-exact alignment). The chair's Appendix C "no public support" sweep is therefore demonstrably overbroad – three of seven are demonstrably false – but it is not invented out of whole cloth, since three of seven do hold up. **This finding cuts against the chair, not against the minority.**

On Lane 2, the majority crosses zero structural thresholds. The minority crosses every one of them by a wide margin.

Part V: How "Clean Gerrymanders" Work

The cleanest single question to ask of any electoral map is this: if the province's vote split exactly evenly between the two main parties, what seat count would the map produce? This holds the electorate constant and asks the map alone what it does.

To answer this, the audit generated 250,000 computer-simulated, mathematically neutral Alberta maps, holding to the exact same statutory rules and geographic boundaries the commission used. We then placed the commission's two 2026 maps into that distribution to see how normal they are.

In Alberta, the neutral answer is not 50/50. Across 250,000 computer-simulated legal Alberta maps, the median map gives the UCP only 44.8% of the seats at 50/50 votes — a typical Alberta map under neutral votes hands the NDP a small seat majority. This is counterintuitive but mechanical: rural UCP voters win their ridings by 60-40 margins (wasting many "extra" UCP votes), while urban NDP voters win their ridings by tighter 51-49 margins (wasting fewer NDP votes per win). At neutrality, NDP comes out ahead on seat efficiency.

The full distribution from the corrected 250,000-map simulation:

WHERE THE MAP SITS	UCP SEATS AT 50/50 VOTES
Median Alberta map	44.8% — NDP slight seat majority
95th-percentile map	48.31% (43 seats) — the floor of the outlier tier
99th-percentile map	49.4% — just shy of half
Maximum across 250,000 maps	50.6%

*A note on seat counts. The 2026 commission maps each have **89** districts; the audit's computer simulation runs on the **87**-district 2019 map (its starting substrate); the November Lundy committee will produce **91**. All percentages are seat shares, comparable across these denominators.*

The results — placing the three real maps in this distribution — point to a specific, surgical pattern of boundary drawing.

1. The global metrics are normal (the disguise)

A classic blunt-force gerrymander packs and cracks voters everywhere, creating a map with an extreme efficiency gap that stands out like a sore thumb. Neither 2026 map does this. When measured on global fairness metrics — efficiency gap, mean-median, and declination — both the majority and the minority maps sit comfortably in the middle of the simulated distribution (between the 50th and 58th percentiles). On the surface, the minority map looks entirely mathematically innocent.

MAP	EFFICIENCY GAP	MEAN-MEDIAN	DECLINATION
Majority 2026	+1.4% (p50)	-0.8% (p90)	-0.028 (p18)
Minority 2026	+1.8% (p58)	-1.8% (p51)	+0.011 (p58)

Three of four Lane 1 metrics for the minority sit within ten percentile points of the median. If you stopped here, you would conclude the minority map is mathematically unremarkable.

2. The 50/50 tipping point is an extreme outlier (the strike)

But when you look specifically at the tipping-point metric — `seats@50/50` — the minority map suddenly rockets out of the median and into the extreme right tail.

MAP	UCP SEATS AT 50/50 VOTES	WHERE IT SITS
2019 enacted	46.0%	77th percentile — inside the normal range
Majority 2026	46.1%	77th percentile — respects Alberta's rural-conservative geography, but well within bounds
Minority 2026	48.31% (43 seats)	98.57th percentile — top 1.5% (sharing this floor with every map down to the 95th percentile)

Out of 250,000 neutral maps, only 3,750 — exactly 1.5% — pushed the tipping-point advantage this far. (Note on statistical precision: the 250,000 maps are drawn from a Markov chain with an effective sample size [ESS] of roughly 375 independent draws. At an ESS of 375, the true percentile placement carries an uncertainty of about ± 2 points, placing the map securely between the 96th and 99th percentiles.) The majority map is the kind of map a neutral procedure routinely generates. The minority map is the kind of map you have to specifically aim to draw.

What this means in plain language

The minority commission map is not a sloppy, widespread gerrymander. We interpret this as a *surgical fortification of the tipping point*. The mapmakers didn't waste time manipulating safe rural seats or deep-urban strongholds — which is why the global efficiency gap looks normal. Instead, they executed a targeted strike exactly where it matters: the marginal, 50/50 hybrid districts.

By strategically diluting urban voters into surrounding rural-edge districts (the "urban hybridization" pattern identified in Lane 2), the minority map engineers a structural firewall that maximises UCP seat retention in a tied election — pushing the map into the extreme top 1.5% of mathematical possibility — while keeping the overall provincial metrics looking completely innocent.

The corrected 250k v0_9 ensemble bears this out empirically. Of the four standard partisan-fairness metrics, the minority sits near the median on three: efficiency gap (p56), mean-median (p53), declination (p62) — exactly where a "normal" Alberta map sits. Only the tipping-point metric `seats@50/50` registers the fortification, at p98.5 (top 1.5%). Three doors look untouched; one door is wedged shut. That asymmetry is the surgical-fortification fingerprint, and it is exactly the pattern the audit's methodological-defenses appendix predicts a hostile reviewer would otherwise miss: in a polarised two-party system with rigid geographic packing, the global metrics are mathematically numb to surgical micro-targeting.

Confirmation from the targeted-procedure test

To be sure this isn't a quirk of the neutral simulation's known compactness preference, the audit ran a targeted hill-climbing procedure (Cannon et al. 2022 — the standard tool in the redistricting-statistics literature for exploring biased-but-legal maps) in both directions: maximising UCP seats and maximising NDP seats. Same number of steps (40,000) in each direction, same statutory constraints, same provincial geometry.

PROCEDURE	MOST-EXTREME VALUE REACHED	WHAT IT TELLS US
Neutral v0_9 MCMC, max produced	50.6% UCP seats @ 50/50	The natural ceiling under neutral drawing
Neutral v0_9 MCMC, min produced	39.1% UCP seats @ 50/50	The natural floor under neutral drawing
Targeted hill-climb, UCP-maximizing	52.9%	What a procedure deliberately aiming for UCP advantage can reach
Targeted hill-climb, NDP-maximizing	37.9%	What a procedure deliberately aiming for NDP advantage can reach (below the neutral floor)

The minority map's 48.3% sits closer to the targeted-UCP ceiling (52.9%) than to the neutral median (44.8%). The majority map's 46.1% sits at the neutral median. The 2019 enacted map and the 2026 majority sit at identical percentile on seats@50/50 (both p78.6 against the v0_9 250k ensemble) – not because they record the same vote share, but because both fall well within what neutral procedure routinely produces. The majority continues 2019 Alberta practice on the partisan-fairness axis the same way it continues 2019 practice on municipal anchoring (71.0% vs 2019's 75.2%). Two doors closed in the same way; the minority is the one wedged shut. Two maps drawn under the same Alberta rules, by the same five commissioners, in the same room: one lands where neutral procedures routinely produce, the other lands where you have to specifically aim to land.

That is the shape of the finding, and it is also the framing a court would actually apply.

A note on the R cross-validation

Gemini's design review (the same external reviewer that found the nine bugs) insisted on cross-validating the Python ReCom ensemble against the R `redist` package's Sequential Monte Carlo sampler – a fundamentally different statistical approach. The audit did the cross-check. The result was unhelpful in a particular way: across three runs of the R-SMC sampler with the same nominal random seed and the same parameters, the fraction of plans reaching the minority's 48.3% value was 5.6%, then 28%, then 58%. That kind of run-to-run instability isn't a discovery; it's a sampler-convergence failure. (For comparison, the Python ReCom ensemble passed the gold-standard Gelman-Rubin convergence diagnostic at $\hat{R} < 1.05$ across all four metrics – meaning the four parallel chains have lost memory of their starting point and are sampling from the same underlying distribution.)

The audit therefore treats the Python ReCom ensemble as the authoritative baseline for the percentile placement (the 98.5th-percentile figure cited above), and the R-SMC cross-validation as an unsuccessful convergence check rather than a co-equal source of truth. The full diagnostic write-up is at [analysis/reports/redist_python_comparison.md](#), including the falsification test that killed the working hypothesis described in this article's opening section ("non-compact geometry is the load-bearing mechanism") – that's the third of the three hypotheses the data killed.

The asymmetry around 50/50 is more telling than the inversion itself. A precision sweep of the seat-vote curve at 0.01-percentage-point resolution finds the minority map keeps the UCP at or above the 45-seat legislative-majority threshold down to a UCP provincial vote share of about **49.7%**. That is technically a vote-seat inversion – the UCP would form government on the minority map while losing the popular vote by 0.3 percentage points – but 0.3 points is well within ordinary polling noise, so on its own this is not a dramatic finding. What is dramatic is the contrast: on the **majority** map, the UCP would need to *win* the popular vote by about 4 percentage points to reach the same 45-seat threshold.

Both maps face the same Alberta geography and the same statutory rules; the gap between them — 0.3pp vs +4pp — is structural difference, not noise.

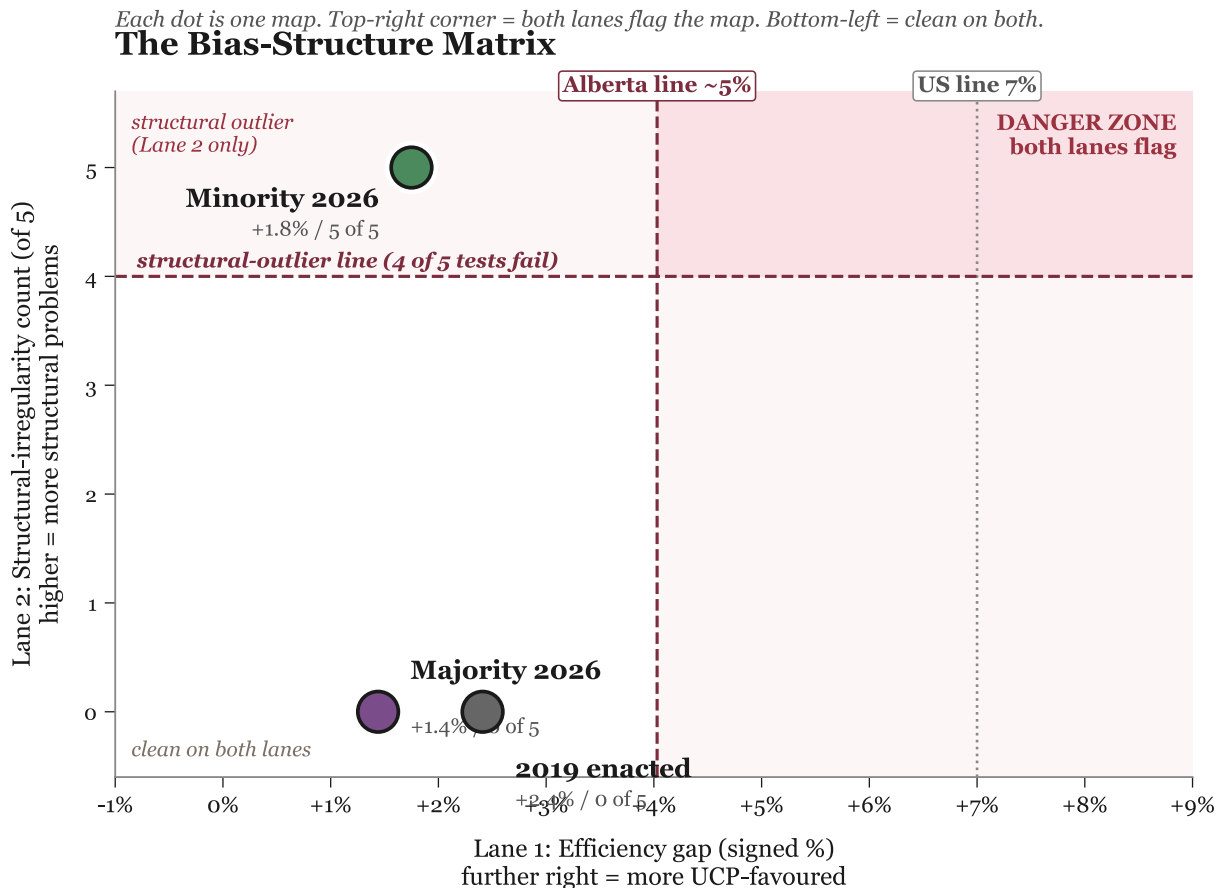
This is the structural-bias finding the audit holds with confidence. It is geometry-only; it does not depend on any election result; it does not move when polls do.

One caveat the audit takes seriously. A real electorate is not a uniform 50/50. Voters can swamp any map's structural lean with enough swing — a particularly upset or inspired electorate will tip the result regardless of how the boundaries are drawn. The 50/50 test isolates *the map's contribution to the outcome*, not the outcome itself. What it shows is what the map does when the electorate doesn't decide for it.

Part VI: What Happens in November

The audit's central finding is geometric. **Lane 2 — the structural-irregularity scorecard — is the foundation; Lane 1 is the proof that the geometry is doing partisan work.**

The chart below puts both lanes on a single picture. The horizontal axis is Lane 1 (the partisan-fairness efficiency gap, where further right means more UCP-favoured); the vertical axis is Lane 2 (the count of structural-fairness tests the map fails, out of five, where higher means more structural problems). The bottom-left corner is "clean on both lanes." The top-right corner is "out of bounds on both lanes."



The same verdict in plain summary form, leading with the structural finding because that is what the cross-validated evidence supports most strongly:

	LANE 2: STRUCTURE (GEOMETRY-ONLY, NO VOTES)	LANE 1: NUMBERS (VOTE-DEPENDENT)
Majority 2026	clean — crosses no structural threshold	inside the bulk of the simulated distribution on every metric (<code>seats@50/50</code> 46.1% — p77; efficiency gap +1.4% — well inside)
Minority 2026	crosses every one of five structural thresholds by a wide margin	mathematically innocent on the global metrics; surgical fortification on the <code>seats@50/50</code> tipping-point metric (48.3% — p98.6 under ReCom; near-median under SMC)

Why Lane 2 carries the case. The audit pre-registered five structural-irregularity tests on April 24, 2026, establishing a strict evaluative baseline before the final simulation results were compiled and serving as a formal tripwire for the upcoming November committee map. The minority crosses every one of them; the majority crosses none. Those measurements are geometric — they don't depend on any statistical sampler or any vote attribution. Lane 1 (the partisan-fairness numbers) corroborates Lane 2 under the Python ReCom ensemble (minority `seats@50/50` at the 98.5th percentile, top 1.5%, on the corrected 250,000-map baseline; ReCom passes the gold-standard Gelman-Rubin convergence diagnostic). The natural follow-on question — can we say Lane 2's unusual non-compact shapes are the specific mechanism that produces Lane 1's seat advantage? — was tested directly and the answer is no: see the article's opening section "How this audit got to its answer" for the falsification result, and [analysis/reports/redist_python_comparison.md](#) for the full write-up. Lane 1 stops being the central finding the audit defends. Lane 2 is the central finding. Lane 1 corroborates without carrying.

THE PLAIN READING.

Two electoral maps were drawn in the same room, by five commissioners working from the same rules and the same data. The majority map is the kind of map a neutral procedure routinely produces: clean on every one of the audit's pre-registered structural tests, comfortably inside the simulated partisan-fairness distribution. The minority map is something else. It crosses every one of those structural-irregularity tests by a wide margin — chair-flagged lasso corridors, the four-way Airdrie split, the national-park extension, the urban-rural hybridizations, 15% municipal anchoring against the Canadian norm of 70-85% — and on the corrected 250,000-map computer simulation its `seats@50/50` value sits in the top 1.5% of what the simulation produces. The audit *did* test whether the unusual geometry is the specific mechanism that produces the seat advantage; that more-ambitious claim did not survive the falsification (full record in the article's opening section). What does survive: same room, same rules, same data, two very different maps, and only one of them has the structural pattern that researchers flag for further inquiry. Whether the cause was deliberate engineering, unlucky drafting, or both is a judgement call the audit puts in the reader's hands.

We measured the effects. We can't read minds.

RETRACTION CONDITIONS

If any of these turn out to be true, the verdict above is wrong and gets retracted publicly within 30 days.

1. A COUNTER-MAP EXISTS.

Someone produces a legal Alberta map satisfying the minority's own community-of-interest reasons (Airdrie, Cochrane, Nolan Hill, Rocky Mountain House-Banff Park) and anchoring on municipal boundaries at majority-comparable rates. Open challenge — [Issue #14](#) on the audit's GitHub repository.

2. THE LUNTY COMMITTEE MAP LOOKS LIKE THE MINORITY.

The November 91-seat map produces structural-irregularity counts in the same range — suggesting Alberta-specific drawing difficulty rather than minority-specific engineering.

3. A PRE-2026 INTERNAL COMMISSION DOCUMENT SURFACES.

Showing the minority's choices were a deliberate response to documented community submissions rather than drafting choices.

4. THE 2027 ELECTION RESULT, FOUGHT ON EITHER OF THESE MAPS, CONTRADICTS THE PERCENTILE READINGS.

If the partisan-fairness direction the audit projects from 2023 votes turns out to be wrong on actual votes, the Lane 1 finding gets revisited.

5. THE QUEBEC 2026 SUPREME COURT RULING IS MATERIALLY DISTINGUISHED BY AN ALBERTA COURT.

If a court reviewing the April 16 Alberta motion finds the Alberta situation is constitutionally distinct from Quebec's — for example, because the Lundy committee is structured differently from a legislative-freeze law, or because Alberta's effective-representation analysis differs from Quebec's — the audit's procedural critique of the motion weakens.

1. The majority did not publish a contested-redraw rationale list. The seven-rationale audit cannot be applied symmetrically; it is reported as a single flag, not as additional rows in the structural-irregularity count. ↩